

Multiple Choice

1. **Answer:** D

Options: A) ExoMars; B) Mars Exploration Rovers; C) Mars Science Laboratory; D) Phoenix Mars Lander

Note: Correct option: D - Phoenix Mars Lander

2. **Answer:** C

Options: A) red dwarfs.; B) pulsars.; C) black holes.; D) galaxies.

Note: Correct option: C - black holes.

3. **Answer:** B

Options: A) nuclear fusion in the core; B) by contracting changing gravitational potential energy into thermal energy; C) internal friction due to its high rotation rate; D) chemical processes

Note: Correct option: B - by contracting changing gravitational potential energy into thermal energy

4. **Answer:** B

Options: A) Planets orbit around the Sun in nearly circular orbits in a flattened disk.; B) the equal number of terrestrial and jovian planets; C) the craters on the Moon; D) asteroids Kuiper-belt comets and the Oort cloud

Note: Correct option: B - the equal number of terrestrial and jovian planets

5. **Answer:** D

Options: A) Will the Sun become a black hole?; B) Is the universe infinitely large?; C) How old is the visible universe?; D) Are we alone in the universe?

Note: Correct option: D - Are we alone in the universe?

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The luminosity integrated over all wavelengths.'

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Asteroids have substantial motion relative to the background stars.'

Long Form Response

11. **Answer:** The Moon split from the Earth due to tidal forces.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Orion. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key